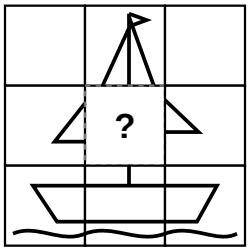
Math Kangaroo Practice - Grades 3-4

Practice set for 2002 #1
10 practice problems per selected problem.

Set 1 - Missing puzzle tile

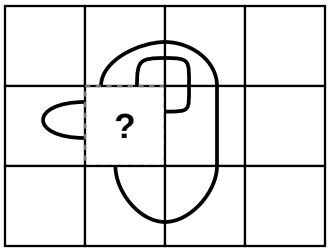
Based on 2002, Problem 1

Practice 1.1 Which of the squares below is missing from the picture of the sailboat?



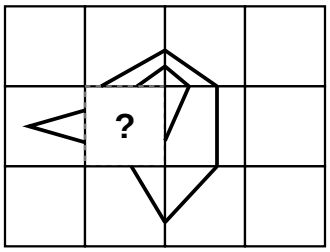
A)
B)
C)
D)
E)

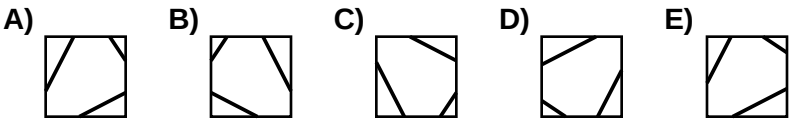
Practice 1.2 Park paths wind across the 3 × 4 map below, and one square tile has been removed. Which of the tiles (A through E) belongs in the square marked with a question mark (?)?



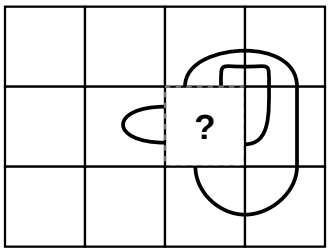
A)
B)
C)
D)
E)

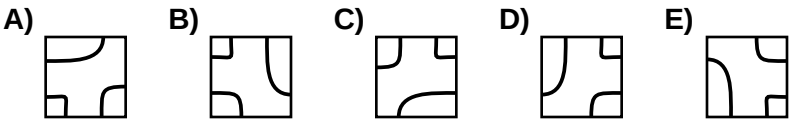
Practice 1.3 One square tile is missing from the 3 × 4 geometric line pattern below. Which tile (A through E) fits in the empty square marked with a question mark (?)?



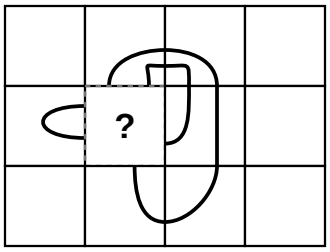


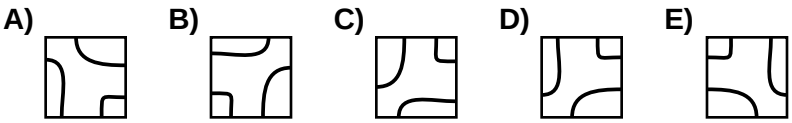
Practice 1.4 A model train track loops across the 3 × 4 board below. Which tile (A through E) belongs in the missing square marked with a question mark (?) to make the track continuous?



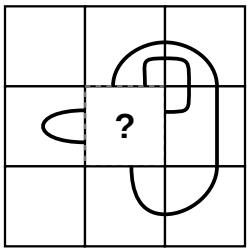


Practice 1.5 Park paths wind across the 3 × 4 map below, and one square tile has been removed. Which of the tiles (A through E) belongs in the square marked with a question mark (?)?



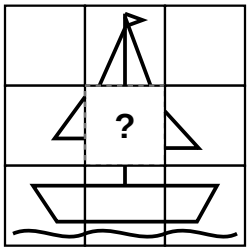


Practice 1.6 A model train track loops across the 3 × 3 board below. Which tile (A through E) belongs in the missing square marked with a question mark (?) to make the track continuous?



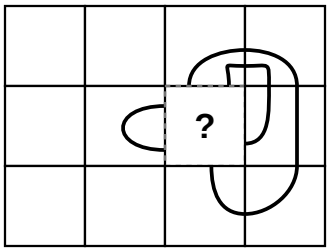
A)
B)
C)
D)
E)

Practice 1.7 Which of the squares below is missing from the picture of the sailboat?



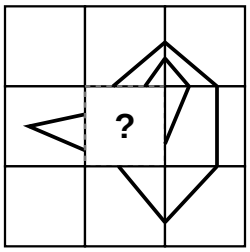
A)
B)
C)
D)
E)

Practice 1.8 Park paths wind across the 3 × 4 map below, and one square tile has been removed. Which of the tiles (A through E) belongs in the square marked with a question mark (?)?



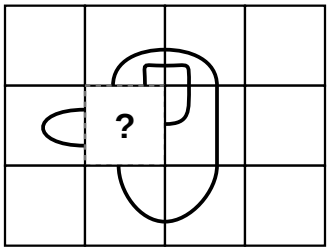
A)
B)
C)
D)
E)

Practice 1.9 One square tile is missing from the 3 × 3 geometric line pattern below. Which tile (A through E) fits in the empty square marked with a question mark (?)?



A)
B)
C)
D)
E)

Practice 1.10 A model train track loops across the 3 × 4 board below. Which tile (A through E) belongs in the missing square marked with a question mark (?) to make the track continuous?



A)
B)
C)
D)
E)

Answer Key & Explanations

For generated/2026-10-19-21-17-18.pdf (Practice set for 2002 #1)

Solutions & kid-friendly explanations

1.1: A In the missing center square, the vertical mast runs from the top (0.55) to the bottom (0.55), the two upper sail edges enter from the top at 0.22 and 0.86, and the horizontal sail booms sit at heights 0.30 (left) and 0.42 (right). Only square A matches all four edges.

1.2: D Checking the four edges of the missing square (?): two lines enter from the top (at 0.20 and 0.65), two enter from the left (at 0.35 and 0.80), one exits through the bottom (at 0.38), and one exits through the right (at 0.68). Only tile D connects all six boundary points in the proper orientation.

1.3: B Checking the four edges of the missing square (?): two lines enter from the top (at 0.20 and 0.65), two enter from the left (at 0.30 and 0.70), one exits through the bottom (at 0.58), and one exits through the right (at 0.32). Only tile B connects all six boundary points in the proper orientation.

1.4: B Checking the four edges of the missing square (?): two lines enter from the top (at 0.25 and 0.70), two enter from the left (at 0.30 and 0.75), one exits through the bottom (at 0.38), and one exits through the right (at 0.28). Only tile B connects all six boundary points in the proper orientation.

1.5: E Checking the four edges of the missing square (?): two lines enter from the top (at 0.30 and 0.80), two enter from the left (at 0.35 and 0.75), one exits through the bottom (at 0.62), and one exits through the right (at 0.28). Only tile E connects all six boundary points in the proper orientation.

1.6: A Checking the four edges of the missing square (?): two lines enter from the top (at 0.35 and 0.75), two enter from the left (at 0.30 and 0.70), one exits through the bottom (at 0.58), and one exits through the right (at 0.68). Only tile A connects all six boundary points in the proper orientation.

1.7: B In the missing center square, the vertical mast runs from the top (0.50) to the bottom (0.50), the two upper sail edges enter from the top at 0.18 and 0.82, and the horizontal sail booms sit at heights 0.34 (left) and 0.22 (right). Only square B matches all four edges.

1.8: A Checking the four edges of the missing square (?): two lines enter from the top (at 0.30 and 0.80), two enter from the left (at 0.20 and 0.75), one exits through the bottom (at 0.58), and one exits through the right (at 0.28). Only tile A connects all six boundary points in the proper orientation.

1.9: A Checking the four edges of the missing square (?): two lines enter from the top (at 0.35 and 0.75), two enter from the left (at 0.20 and 0.65), one exits through the bottom (at 0.42), and one exits through the right (at 0.28). Only tile A connects all six boundary points in the proper orientation.

1.10: B Checking the four edges of the missing square (?): two lines enter from the top (at 0.35 and 0.75), two enter from the left (at 0.25 and 0.70), one exits through the bottom (at 0.42), and one exits through the right (at 0.52). Only tile B connects all six boundary points in the proper orientation.